

THE SUN

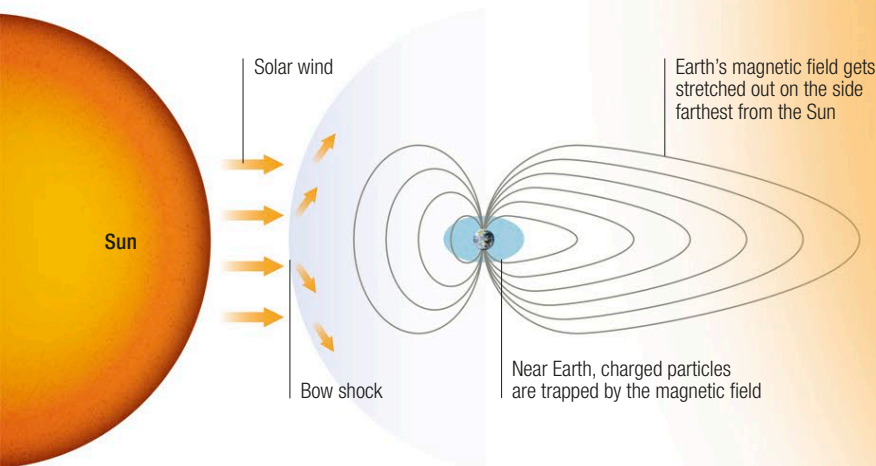
The upper layers of the Sun's atmosphere, the corona, is far hotter than the Sun's surface

ESSENTIAL TO LIFE ON EARTH BY PROVIDING US WITH WARMTH AND LIGHT, THE SUN MAY SEEM SPECIAL, BUT IT IS JUST A TYPICAL STAR. AS IT IS SO CLOSE TO US, IT HAS BEEN STUDIED IN GREAT DETAIL.

Measuring 863,000 miles (1.39 million km) across and rotating every 24.5 days, the Sun is a nuclear fusion furnace in which atoms are crushed together in its core, releasing vast quantities of heat and light. The surface that we see, at 10,800°F (6,000°C), is only part of a complex, seething jumble of magnetic field and charged particles. Almost everything we can see on the Sun is in a state of matter called a plasma, where gas has separated into negatively charged electrons and positively charged ions (atoms or molecules that have lost electrons). The Sun has an activity cycle of around 11 years, during which the numbers of sunspots, flares, and eruptions rise and fall significantly.

The solar wind

A flow of charged particles, called the solar wind, flows continuously from the Sun into space at hundreds of miles per second. This wind carves out an enormous bubble in space called the heliosphere; Earth and all the other planets orbit within it, shielded from interstellar space. Like weather on the Earth, the solar wind is variable. It can be gusty, and its effects on planets and comets can change abruptly.



△ Earth's magnetosphere

Our planet is protected from the solar wind by its magnetic field. This magnetic bubble, called the magnetosphere, allows the solar wind to usually only affect Earth near the poles, where the northern and southern lights—the aurorae—are seen.

▷ Saturn's aurora

Like Earth and most other planets, Saturn has a magnetosphere. The interaction between the solar wind and the planet's magnetic field generates aurora near its poles, as occurs on our planet. Here, ultraviolet light shows the southern lights near the South Pole.

